

MATH 1210 Worksheet 3 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

<http://home.cc.umanitoba.ca/~thomas/Courses>

1. Write the following in Cartesian form.

(a) $e^{-i\pi/2}$

(b) $4 \frac{e^{2i\pi/3}}{e^{i\pi/6}}$

Solution:

(a) $e^{-i\pi/2} = -i$

(b) $4 \frac{e^{2i\pi/3}}{e^{i\pi/6}} = 4e^{2i\pi/3-i\pi/6} = 4e^{4i\pi/6-i\pi/6} = 4e^{3i\pi/6} = 4e^{i\pi/2} = 4i.$

2. Write

$$\overline{(2 + \sqrt{2}e^{i5\pi/4})} \cdot \frac{1}{1 - e^{i\pi/3}}$$

in exponential form. Use the principal value of the argument.

Solution: We have $e^{i5\pi/4} = -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$ and $e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. So

$$2 + \sqrt{2}e^{i5\pi/4} = 2 - 1 - i = 1 - i = \sqrt{2}e^{-i\pi/4},$$

and the numerator is

$$\overline{2 + \sqrt{2}e^{i5\pi/4}} = \overline{\sqrt{2}e^{-i\pi/4}} = \sqrt{2}e^{i\pi/4}.$$

For the denominator we find

$$1 - e^{i\pi/3} = 1 - \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = \frac{1}{2} - i\frac{\sqrt{3}}{2} = e^{-i\pi/3}.$$

Hence, we find that

$$\frac{\overline{2 + \sqrt{2}e^{i5\pi/4}}}{1 - e^{i\pi/3}} = \frac{\sqrt{2}e^{i\pi/4}}{e^{-i\pi/3}} = \sqrt{2}e^{i\pi/4}e^{i\pi/3} = \sqrt{2}e^{i\pi(\frac{1}{4}+\frac{1}{3})} = \sqrt{2}e^{i7\pi/12}.$$

3. Find all the complex numbers z solving the equation

$$z^5 = -16\sqrt{2} - 16\sqrt{2}i.$$

Write them in exponential form and use principal value of their arguments.

Solution: We write the right-hand side in exponential form:

$$-16\sqrt{2} - 16\sqrt{2}i = 32 \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right) = 2^5 e^{-i3\pi/4}.$$

Let us set $z = re^{i\theta}$. If z is a solution of

$$z^5 = -16\sqrt{2} - 16\sqrt{2}i = 2^5 e^{-i3\pi/4}. \quad (1)$$

then

$$r^5 e^{i5\theta} = 2^5 e^{-i3\pi/4}.$$

This implies that $r = 2$ and

$$\begin{aligned} 5\theta &= -3\pi/4 + 2\pi k = (-3 + 8k)\pi/4 \\ \iff \theta &= (-3 + 8k)\pi/20 \end{aligned}$$

where k is an integer. Hence z must have the form $z = 2e^{(-3+8k)i\pi/20}$. For $k = 0, 1, 2, 3, 4$ we get the five solutions to the equation (1)

$$\begin{aligned} k=0, \quad \theta &= -3\pi/20, \quad z_0 = 2e^{-3i\pi/20} \\ k=1, \quad \theta &= 5\pi/20, \quad z_1 = 2e^{5i\pi/20} \\ k=2, \quad \theta &= 13\pi/20, \quad z_2 = 2e^{13i\pi/20} \\ k=3, \quad \theta &= 21\pi/20, \quad z_3 = 2e^{21i\pi/20} = 2e^{-19i\pi/20} \\ k=4, \quad \theta &= 29\pi/20, \quad z_4 = 2e^{29i\pi/20} = 2e^{-11i\pi/20}. \end{aligned}$$

4. Find all the solution of the equation $x^6 - 4x^3 + 8 = 0$. Write the roots in exponential form and use the principal value of their arguments.

Solution: If we write $z = x^3$, then the equation becomes $z^2 - 4z + 8 = 0$. It is a quadratic equation that has two solutions are

$$z = \frac{4 \pm \sqrt{4^2 - 4 \cdot 8}}{2} = 2 \pm 2i.$$

So we have to find all the complex numbers x such that $x^3 = 2 + 2i$ or $x^3 = 2 - 2i$.

- First let us solve $x^3 = 2 + 2i$. In exponential form the right-hand side is

$$2 + 2i = 2\sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i \right) = \sqrt{2}^3 e^{i\pi/4}.$$

Hence the equation $x^3 = 2 + 2i$ can be written

$$x^3 = \sqrt{2}^3 e^{i\pi/4}.$$

Let us set $x = re^{i\theta}$. If x is a solution of

$$x^3 = \sqrt{2}^3 e^{i\pi/4}. \quad (2)$$

then

$$r^3 e^{i3\theta} = \sqrt{2}^3 e^{i\pi/4}.$$

This implies that $r = \sqrt{2}$ and

$$\begin{aligned} 3\theta &= i\pi/4 + 2\pi k = (1 + 8k)\pi/4 \\ \iff \theta &= (1 + 8k)\pi/12 \end{aligned}$$

where k is an integer. Hence x must have the form $x = \sqrt{2}(1 + 8k)\pi/12$. The principal values of the argument are obtained for $k = 0, 1, 2$, we get the five solutions to the equation (2)

$$\sqrt{2}e^{i\pi/12}, \quad \sqrt{2}e^{i9\pi/12}, \quad \text{and} \quad \sqrt{2}e^{17i\pi/12} = \sqrt{2}e^{-7i\pi/12}. \quad (3)$$

- For the equation $x^3 = 2 - 2i$ we can reduce to the first case. Indeed, if x satisfies $x^3 = 2 + 2i$ then the complex conjugate satisfies $\bar{x}^3 = 2 - 2i$. Hence the solutions of $x^3 = 2 - 2i$ are the complex conjugates of the solutions found in (3):

$$\sqrt{2}e^{-i\pi/12}, \quad \sqrt{2}e^{-i9\pi/12}, \quad \text{and} \quad \sqrt{2}e^{7i\pi/12}. \quad (4)$$

In conclusion, we find that the 6 roots of the equation $x^6 - 4x^3 + 8 = 0$ are:

$$\sqrt{2}e^{\pm i\pi/12}, \quad \sqrt{2}e^{\pm i9\pi/12}, \quad \text{and} \quad \sqrt{2}e^{\pm i7\pi/12}. \quad (5)$$

5. Use polynomial division to divide $P(x) = 7x^5 - 11x^4 - 8x^3 + 9x^2 - 13x + 6$ by $x - 2$. Deduce that 2 is a root.

Solution: The polynomial division of $P(x)$ by $x - 2$ is

$$\begin{array}{r}
 7x^4 + 3x^3 - 2x^2 + 5x - 3 \\
x-2 \overline{) 7x^5 - 11x^4 - 8x^3 + 9x^2 - 13x + 6} \\
\underline{-7x^5 + 14x^4} \\
 3x^4 - 8x^3 \\
\underline{-3x^4 + 6x^3} \\
 -2x^3 + 9x^2 \\
 \underline{2x^3 - 4x^2} \\
 5x^2 - 13x \\
 \underline{-5x^2 + 10x} \\
 -3x + 6 \\
 \underline{3x - 6} \\
 0
\end{array}$$

Hence

$$7x^5 - 11x^4 - 8x^3 + 9x^2 - 13x + 6 = (x - 2)(7x^4 + 3x^3 - 2x^2 + 5x - 3).$$

By the factor theorem, 2 is a root since $x - 2$ is a factor. .

6. Use polynomial division to divide $P(x) = x^4 + x^3 + 3x^2 - x + 1$ by $x + 1$. Deduce that -1 is not a root.

Solution: The polynomial division of $P(x)$ by $x + 1$ is

$$\begin{array}{r}
 x^3 \quad + 3x - 4 \\
 x+1 \overline{) \quad x^4 + x^3 + 3x^2 - x + 1} \\
 \underline{-x^4 - x^3} \\
 3x^2 - x \\
 \underline{-3x^2 - 3x} \\
 -4x + 1 \\
 \underline{4x + 4} \\
 5
 \end{array}$$

Hence

$$x^4 + x^3 + 3x^2 - x + 1 = (x+1)(x^3 + 3x - 4) + 5.$$

We have $P(-1) = 5$ by the remainder theorem, -1 is not a root .

7. If $-2 + 3i$ is a zero of the polynomial $P(x) = x^3 + 7x^2 + 25x + 39$, find its other zeros.

Solution: If $-2 + 3i$ is a root of the polynomial, then $-2 - 3i$ is a root as well since the polynomial has real coefficients. Hence $P(x)$ is divisible by

$$(x - (-2 + 3i))(x - (-2 - 3i)) = x^2 + 4x + 13.$$

We divide $P(x)$ by this polynomial:

$$\begin{array}{r}
 x + 3 \\
 x^2 + 4x + 13 \overline{) \quad x^3 + 7x^2 + 25x + 39} \\
 \underline{-x^3 - 4x^2 - 13x} \\
 3x^2 + 12x + 39 \\
 \underline{-3x^2 - 12x - 39} \\
 0
 \end{array}$$

and conclude that

$$P(x) = (x^2 + 4x + 13)(x + 3).$$

It follows that the zeroes of $P(x)$ are

$$-2 + 3i, \quad -2 - 3i \quad \text{and} \quad -3.$$

8. For what value(s) of k will the polynomial $P(x) = 5x^5 + 4x^4 + kx^3 + 2x^2 + x + 1$ have remainder 15 when divided by $x - 1$.

Solution: The remainder of $P(x)$ divided by $x - 1$ is $P(1) = 13 + k$. If we want it to have remainder 15 then k must be 2.

9. Find the remainder when $P(x) = x^{999} + 2x^{998} + x^2 - 1$ is divided by $x + 1$.

Solution: We write $P(x) = (x+1)Q(x) + r$, where r is the remainder. By evaluating both sides at -1 we find that the remainder is $r = P(-1) = 1$.

10. Consider the polynomial

$$P(x) = 4x^5 + 3x^4 - 2x^3 + x^2 - x - 2.$$

- (a) Use Descartes' rule of signs to state the number of possible positive and negative solutions.
- (b) Use Bounds theorem find a bound on the roots.
- (c) Use the Rational Root Theorem to list all possible rational solutions. Use the results from the first two parts.

Solution:

- (a) There are 3 sign changes in $P(x)$, hence there are 1 or 3 positive real solutions. On the other hand, the polynomial $P(-x) = -4x^5 + 3x^4 + 2x^3 + x^2 + x - 2$ has 2 sign changes, so it has 0 or 2 negative real roots.

- (b) We have $M = \max(|3|, |-2|, |1|, |-2|) = 3$. Hence Bounds Theorem tells us that if x is any root (complex or real), then

$$|x| < \frac{M}{|a_5|} + 1 = \frac{3}{4} + 1 = \frac{7}{4}.$$

- (c) By Bounds theorem, if $x = p/q$ is a rational solution then p divides 2 and q divides 4. Hence p can be $\pm 1, \pm 2$ and q can be $\pm 1, \pm 2, \pm 4$. Hence the possible rational roots are

$$x = \pm 1/4, \pm 1/2, \pm 1, \pm 2.$$

By Bounds theorem we can eliminate ± 2 since $|-2| \geq \frac{7}{4}$ does not satisfy the bound of Bounds theorem. Then, the possible rational roots are

$$x = \pm 1/4, \pm 1/2, \pm 1.$$

11. Consider the polynomial

$$P(x) = x^3 + 4x^2 + 12x + 9.$$

- (a) Show that it has no zeroes in the interval $[-100, -20]$.
- (b) Show that it has no zeroes in the interval $[0, 13]$.

Solution:

- (a) We apply Bounds Theorem. We have $M = \max(|4|, |12|, |9|) = 12$. Hence Bounds Theorem tells us that if x is a root, then

$$|x| < \frac{M}{|a_3|} + 1 = 12 + 1 = 13.$$

In particular, any number x in the interval $[-100, -20]$ satisfy $|x| \geq 20 > 13$, so cannot be a root.

- (b) There are 0 sign changes in $P(x)$, hence there are no positive real solutions. In particular this imply that there are no solutions in $(0, 13]$. We can then check that at 0 we have $P(0) = 9 \neq 0$.

12. Consider the polynomial

$$P(x) = 2x^4 + 2x^3 + x^2 + 5x - 10.$$

- (a) Use the Rational Root Theorem to determine all possible rational roots.
- (b) Let $Q(x) = (x - 5)P(x)$. How many positive real roots and negative real roots does $Q(x)$ have? Explain.
- (c) Use Bounds Theorem to determine an upper bound for the modulus of roots of $P(x)$. Does this eliminate any possible rational roots? Which ones?

Solution:

- (a) If $x = p/q$ is a rational solution then p divides 10 and q divides 2. Hence p can be $\pm 1, \pm 2, \pm 5, \pm 10$ and q can be $\pm 1, \pm 2$. Hence the possible rational roots are

$$x = \pm 1/2, \pm 1, \pm 2, \pm 5/2 \pm 5, \pm 10.$$

- (b) There is 1 sign change in $P(x)$, hence there is 1 positive real solution. On the other hand, the polynomial $P(-x) = 2x^4 - 2x^3 + x^2 - 5x - 10$ has three sign changes, so it has 1 or 3 negative roots. The polynomial $Q(x)$ has the same roots as $P(x)$, plus the (positive) root $x = 5$. Hence $Q(x)$ has 2 positive roots, and 1 or 3 negative roots.

- (c) We have $M = \max(|-10|, |5|, |1|, |2|) = 10$. Bounds Theorem tells us that if x is any root (complex or real), then

$$|x| < \frac{M}{|a_4|} + 1 = \frac{10}{2} + 1 = 6.$$

Hence, we can eliminate the possible rational roots ± 10 .

13. Consider the polynomial

$$g(x) = x^3 - mx^2 - 1$$

where m is an integer. Find all values of m for which $g(x)$ has a rational root.

Solution: By Rational root theorem, if x is a rational solution then $x = \pm 1$. When $x = 1$, we see that $g(x)$ has a root when $m = 0$. When $x = -1$, we see that $g(x)$ has a root when $m = -2$. Hence the possible values are $m = 0$ and $m = -2$.